RED SEAL EXAM PREP

310T Truck & Transport Mechanic

Free 50-Question Mock Exam

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Mock Exam — answer all questions, then check the key

Q1. **A truck automatic transmission shifts late (high RPM shifts) in all ranges. The fluid is at proper level and the correct colour. What is the MOST likely cause?**

A) The throttle position sensor reads lower than actual demand

B) Low line pressure causing slippage that delays shift completion

C) A clogged transmission filter reducing flow to the valve body

D) The throttle position sensor reads higher than actual demand

Q2. **During clutch replacement the flywheel friction face shows light heat checking and shallow grooving, and measurement shows the face can be cleaned up and still finish above the minimum thickness marked on the flywheel. What is the correct action?**

A) Recondition the face, then recheck thickness and runout before installing the clutch

B) Hand-sand the face with abrasive paper until the checking no longer shows

C) Fill the checked area with clutch adhesive so the new disc beds on a smooth surface

D) Install the new clutch on the face as found — the friction material will conform to it

Q3. **During a pre-trip air brake test, a driver finds the air pressure drops more than 3 PSI per minute with the engine off and brakes released. What does this indicate?**

A) Excessive system air leakage that must be located and repaired

B) Normal — 3 PSI per minute is within acceptable leakage limits

C) The air dryer purge valve is open — close it to stop the leak

D) The governor is faulty — it should hold pressure with the engine off

Q4. **A truck driver reports a "chatter" or vibration in the driveline only during low-speed tight turns. The MOST likely cause is:**

A) Worn steering linkage

B) Worn universal joints

C) Loose engine mounts

D) Differential binding

Q5. **Under NSC Standard 11, at a periodic inspection a truck's air brake system is rejected if, during the air leakage test (engine off, service brakes released), pressure drops faster than:**

A) 69 kPa (10 psi) per minute

B) 41 kPa (6 psi) per minute

C) 20 kPa (3 psi) per minute

D) 7 kPa (1 psi) per minute

Q6. **A truck driver reports a vibration that occurs only at highway speeds (above 90 km/h) and goes away below 70 km/h. The vibration is felt through the floor, not the steering wheel. What is the most likely cause?**

A) Engine misfire — engine misfires create vibration through the body at high RPM only

B) Driveshaft imbalance or phasing error — felt through the floor within a specific speed range

C) Front wheel imbalance — imbalance is always felt through the steering wheel

D) Tire flat spot from hard braking — flat spots occur at any speed, not only above 90 km/h

Q7. **A truck's headlights are dim and the alternator output is 13.9V. Battery load test shows battery is good. What is the MOST likely cause of dim lights?**

A) The wrong bulb wattage was installed

B) The battery voltage is higher than normal

C) The alternator is undercharging the battery

D) High resistance in the headlight circuit

Q8. **At a periodic commercial motor vehicle inspection under National Safety Code Standard 11, a trailer tire shows a smooth bulge 6 mm high over a sidewall repair, marked with a blue triangular label. How is that tire recorded?**

A) Rejected; the bulge is evidence of casing separation

B) Accepted; a repair bulge is allowed up to 9 mm high

C) Rejected; the blue label marks a plug repair in the sidewall

D) Hazardous condition; the unit must not leave the shop

Q9. **A truck brake drum of 420 mm nominal size carries no wear limit marking, and no OEM or industry figure is available. Under National Safety Code Standard 11, the drum is rejected once its measured diameter exceeds the original by more than:**

A) 2.3 mm more than the original drum diameter

B) 0.5 mm more than the original drum diameter

C) 3.0 mm more than the original drum diameter

D) 7.5 mm more than the original drum diameter

Q10. **A drive axle is fitted with a limited-slip differential. What does its oil need that a plain differential's oil does not?**

A) A detergent package to keep the clutches clean

B) A higher viscosity oil than the plain axle uses

C) A GL-4 rating instead of the usual GL-5 rating

D) A friction modifier additive for the clutch pack

Q11. **A truck diesel engine has low oil pressure at idle but normal pressure at high RPM. The MOST likely cause is:**

A) Worn main or rod bearings

B) Blocked oil filter

C) Oil viscosity too high

D) Faulty oil pressure sending unit

Q12. **On a truck whose air compressor is gear-driven from the engine, what happens at the compressor when the governor reaches cut-out pressure?**

A) Governor air holds the inlet valves open and the compressor pumps unloaded

B) A clutch in the compressor drive disengages and the compressor stops turning entirely

C) The governor closes a valve in the discharge line so air cannot reach the reservoirs

D) The safety valve on the supply reservoir lifts and vents the excess air to atmosphere

Q13. **A late-model diesel engine logs a fault for "NOx sensor downstream of SCR catalyst reading higher than expected." What is the MOST likely cause?**

A) DEF quality or concentration is out of spec, cutting SCR efficiency

B) The DPF is loaded — soot backpressure is causing incomplete combustion

C) The NOx sensor itself is failed high — replace sensor immediately

D) The EGR valve is stuck open, routing excess exhaust into the intake

Q14. **A tractor has a 4S-4M air-brake ABS system with a pressure modulator valve (PMV) in the service line to each monitored wheel. During an antilock stop, what does the modulator valve do?**

A) It reads wheel rotation and signals the ECU when that wheel starts to lock

B) It applies the spring brake chamber when the ECU detects a locked wheel

C) On ECU command, it exhausts, holds or reapplies service chamber air

D) It proportions front and rear brake pressure to suit the axle loading

Q15. **A truck's battery isolator switch is turned off overnight but the battery is still discharged by morning. The MOST likely cause is:**

A) Normal self-discharge of the battery over 8 hours

B) The starter motor is drawing current continuously

C) The isolator switch is not fully isolating the battery

D) The alternator is draining the battery when engine is off

Q16. **A truck's drum brakes have had a documented full inspection with the drums removed. Under National Safety Code Standard 11 Part B, for how long may its later brake inspections be limited to a check through the inspection holes?**

A) 24 months

B) 7 months

C) 12 months

D) 19 months

Q17. **A flywheel removed during clutch replacement has radial heat cracks running out to the edge of the friction face, and the face already measures at the minimum thickness marked on the flywheel. What is the correct action?**

A) Grind the sharp crack edges smooth with a die grinder and install the new clutch on the face

B) Replace the flywheel — there is no material left to cut and the cracks would remain

C) Heat the face with a torch and let it cool slowly so the cracks close before the clutch goes on

D) Recondition the face on a flywheel grinder — machining takes the cracks out and restores it

Q18. **A truck engine with selective catalytic reduction (SCR) has a warning that DEF quality is poor. The DEF tank was recently filled from a new container. What could cause this DEF quality fault?**

A) DEF is only stable for 6 months — the container was old even if unopened

B) The SCR catalyst is saturated — DEF quality fault appears when the catalyst needs replacement

C) DEF quality sensors always fail after 100,000 km and must be replaced — this is a normal maintenance item

D) Off-spec urea concentration, contamination, or a faulty DEF quality sensor

Q19. **An air brake system governor is set at 125 PSI cut-out and 100 PSI cut-in. A technician resets the governor so that cut-out occurs at 175 PSI. What is the effect on system components?**

A) Pressure beyond what the dryer, valves, fittings and reservoirs are rated for

B) No change — every component in an air brake system is rated to at least 175 PSI

C) Stronger parking brake hold — the spring brakes clamp harder at higher pressure

D) The compressor cycles more often because cut-out now sits further from cut-in

Q20. **Under NSC (National Safety Code) Standard 11B, how often must commercial trucks in Canada undergo a scheduled inspection?**

A) Annually — a full periodic inspection every 12 months

B) Only when the vehicle shows defects — no fixed schedule required

C) Every 25,000 km or 3 months, whichever comes first

D) Once every 5 years — the same as passenger vehicle inspections

Q21. **A truck's diesel engine runs rough at idle, produces white smoke, and has a coolant level that keeps dropping with no visible external leaks. What is the most likely cause?**

A) A failed head gasket letting coolant into the combustion chambers

B) Glow plugs failing — causing hard starts and rough idle when cold

C) Dirty fuel injectors causing incomplete combustion and misfires

D) EGR valve stuck open, flooding cylinders with recirculated exhaust

Q22. **What does "rated RPM" mean for a truck diesel engine?**

A) The idle RPM programmed into the ECM for fuel economy

B) The governor cut-off RPM that prevents engine over-speed

C) The RPM at which the engine produces maximum torque

D) The full-load speed at which maximum power is made

Q23. **On the standard four-terminal mini ISO relay used throughout truck electrical systems, which two terminals connect to the relay's coil - the control side that a dash switch energizes?**

A) Terminals 85 and 87

B) Terminals 85 and 86

C) Terminals 30 and 86

D) Terminals 30 and 87

Q24. **A truck body module controls the exterior lighting with PWM outputs. The headlights dim randomly. A DMM on the headlight feed shows a steady average voltage that does not change when the lights dim. What tool would best diagnose the intermittent signal fault?**

A) DMM set to Hz (frequency) mode

B) A 12-volt incandescent test light

C) An oscilloscope on the PWM output

D) Amp clamp around the headlight wire

Q25. **A truck clutch is slipping — engine RPM rises sharply during acceleration but vehicle speed does not increase proportionally. What would cause clutch slippage?**

A) Clutch disc too thick — excess disc material increases friction surface contact

B) Oil or grease contamination on the clutch disc friction surfaces from a leaking rear main seal

C) Excessive clutch free travel — too much free play causes the pressure plate to apply too much force

D) Incorrect flywheel surface roughness — a smooth flywheel increases friction and causes slippage

Q26. **Following a foundation brake overhaul, a technician sets the initial adjustment on a newly fitted automatic slack adjuster by the manufacturer's installation procedure, then backs it off to confirm lining-to-drum clearance. The adjuster has not taken up on the first four or five brake applications. What is the correct assessment?**

A) Normal — ASAs adjust gradually over multiple applications

B) The ASA pawl is broken — it should adjust on the first application

C) The brake linings are worn out and preventing proper adjustment

D) The adjuster is defective and must be replaced right away

Q27. **A diesel engine's EGR (Exhaust Gas Recirculation) valve is stuck in the open position. What symptoms would this cause at highway cruise speeds?**

A) Engine overheating — recirculated exhaust gases are very hot, increasing coolant temperature

B) Turbocharger surge — excess flow from the EGR valve exceeds turbocharger capacity

C) Rough running, loss of power, and black smoke as oxygen-depleted exhaust dilutes the intake charge

D) No symptoms at cruise — EGR only operates at idle and low loads, not at highway speeds

Q28. **A turbocharged diesel engine has an intercooler. The engine produces less power than specified and intake temperature is higher than normal. A pressure test of the intercooler and charge air piping shows the system holds pressure. What else should be checked?**

A) The intercooler is performing correctly — high intake temperature with no leaks means the turbocharger is the problem

B) Check intercooler mounting — vibration loosens the intercooler and reduces its effectiveness

C) Increase turbocharger boost pressure — the intercooler efficiency is being overcome by insufficient initial boost

D) Perform a flow restriction test across the intercooler — it may hold pressure yet transfer heat poorly

Q29. **A turbocharged diesel engine's MAP (Manifold Absolute Pressure) sensor reads 101 kPa at key-on with the engine stopped at sea level. Under full load the sensor reads 98 kPa. What does this indicate about the MAP sensor and engine?**

A) An intake restriction or boost failure — a turbocharged engine's MAP should be well above atmospheric

B) Normal operation — 98 kPa under full load indicates maximum airflow into the engine

C) Normal for a naturally aspirated engine — manifold pressure stays just below atmospheric under load

D) The MAP sensor is faulty — every diesel engine reads far above atmospheric pressure when fully loaded

Q30. **A driver finds an illuminated amber ABS warning lamp on the trailer during a pre-trip inspection. Under NSC regulations, what action is required?**

A) The driver must reduce maximum speed to 80 km/h until the lamp goes out

B) The trailer must be disconnected — only the tractor may run without ABS

C) The vehicle must be parked immediately — the amber lamp is a critical failure

D) The trip may proceed — record the defect and have it repaired

Q31. **What is the primary difference between a PTC (Positive Temperature Coefficient) thermistor and an NTC (Negative Temperature Coefficient) thermistor in heavy vehicle circuits?**

A) PTC and NTC thermistors are identical — the designation refers only to manufacturing brand

B) PTC is used in AC circuits only. NTC is used in DC circuits only.

C) PTC produces voltage as temperature rises. NTC produces current flow as temperature rises.

D) PTC resistance increases with temperature; NTC resistance decreases with temperature

Q32. **What is the purpose of the two-way check valve (double check valve) in an air brake system?**

A) Allows air to flow in both directions at once for dual-circuit charging

B) Selects the higher of two inlet pressures and passes it to the outlet

C) Prevents air flowing backward from the reservoirs to the compressor

D) Reduces primary-to-secondary pressure to balance brake application

Q33. **At a periodic inspection, a bonded continuous strip drum brake lining on an air-braked truck measures 4 mm at its thinnest point. Under National Safety Code Standard 11, this reading is:**

A) A pass — 1.6 mm is the Canadian minimum for a drum brake shoe lining

B) A pass — only a reading at the centre of the shoe can reject a lining

C) A pass — a lining is a reject only once metal-to-metal contact is heard

D) A reject — continuous strip lining must be at least 5 mm at every point

Q34. **A circuit protected by a 10-amp fuse keeps blowing the fuse. The CORRECT diagnostic approach is:**

A) Install a 20-amp fuse to stop the blowing

B) Add a second 10-amp fuse in parallel

C) Replace the fuse with a circuit breaker of any rating

D) Find and repair the cause of the overcurrent

Q35. **Why does a technician check the piping and clamps between the air cleaner and the turbocharger for leaks?**

A) Fuel pulled in through the leak washes lubricating oil off the cylinder walls

B) Dust drawn in through the leak bypasses the filter and abrades the engine

C) Air drawn in through the leak cools the turbocharger bearings too quickly

D) Air escaping through the leak raises exhaust back pressure at the turbine

Q36. **A Class 8 highway tractor has a bobtail proportioning system in its rear-axle service brake circuit. When the tractor runs bobtail (with no trailer coupled), what does it do?**

A) It sends the unused trailer-circuit air to boost the tractor reservoirs

B) It raises rear brake pressure to make up for the missing trailer brakes

C) It reduces drive-axle brake pressure because the axle is lightly loaded

D) It shuts the rear service brakes off so the steer axle does the stopping

Q37. **A Class 8 tractor-trailer fails a pre-trip inspection because the spring brake parking function will not hold the vehicle stationary on a grade. The reservoirs are full and the pushrod strokes are inside the adjustment limit. Which component failure MOST likely explains this?**

A) A broken power spring in a spring brake chamber, which cannot develop full clamping force

B) The tractor protection valve is closed, cutting off air to the trailer

C) The service brake relay valve is not releasing service air from the chambers

D) The low air pressure warning switch has failed, so the spring brakes were never signalled to apply

Q38. **What type of lubricant is typically required for a truck's manual transmission versus its differentials, and why can't they use the same oil?**

A) Both use the same SAE 80W-90 GL-5 gear oil — there is no difference in lubrication requirements

B) Transmissions typically require GL-4 or a specified fluid; differentials use GL-5, whose EP additives corrode brass synchronizers

C) Both use ATF (Automatic Transmission Fluid) — manual transmissions require ATF for proper synchronizer function

D) Transmissions use engine oil (SAE 30) for its detergent properties. Differentials use gear oil for its extreme pressure additives.

Q39. **What is the function of the quick-release valve in an air brake system?**

A) Quickly increases brake application pressure for emergency stops

B) Allows air to bypass the chambers when the parking brake is engaged

C) Regulates maximum brake application pressure to prevent lockup

D) Rapidly exhausts chamber air when the brakes are released

Q40. **A truck's alternator output voltage is measured at 12.1 volts at idle with minimal electrical load. What does this indicate?**

A) The alternator is functioning correctly — 12.1 volts is the normal charging voltage

B) 12.1 volts is only acceptable at idle — voltage increases to normal under load

C) The alternator is overcharging — voltage above 12 volts will damage batteries

D) The alternator is undercharging — correct charging voltage should be 13.8–14.5 volts for a 12V system

Q41. **An engine oil cooler bypass valve opens when:**

A) Oil is cold and the pressure differential is high

B) The oil pump output exceeds system pressure

C) Oil temperature exceeds the maximum limit

D) Coolant temperature exceeds 95°C

Q42. **Under Canadian regulations, what is required when transporting dangerous goods (DG) by road?**

A) A dangerous goods endorsement added to the driver's licence

B) Training, shipping documents, and vehicle placarding

C) Shipping documents and placards, but training only for the shipper

D) Placards on the vehicle and a training certificate, but no shipping document

Q43. **A relay valve in an air brake system is located at the rear axle brakes. Its purpose is to:**

A) Speed brake application using air from a local reservoir

B) Automatically adjust rear brake pressure based on load

C) Reduce maximum brake pressure to the rear axle

D) Prevent brake pressure from exceeding governor cut-out pressure

Q44. **A truck's alternator is producing 14.2V but the battery is not fully charging after long trips. What should be suspected?**

A) The alternator voltage is too high — it is overcharging the battery

B) The voltage regulator is stuck in the low-voltage position

C) High resistance in the charging circuit or a failing battery

D) The battery needs distilled water — low electrolyte prevents charging

Q45. **What is the driver's responsibility for a vehicle defect found during the daily trip inspection?**

A) Record only a major defect; a minor one needs no written entry at all

B) Note it in the daily log book and keep driving the vehicle as normal

C) Enter it on the trip inspection report; a major defect stops the trip

D) Take the vehicle to a dealer within seven days of finding the defect

Q46. **What voltage is standard for most North American heavy truck electrical systems?**

A) 24V

B) 48V

C) 6V

D) 12V

Q47. **A medium-duty truck is fitted with a synchronized manual transmission, and a technician finds the synchronizer rings worn. What do synchronizers do?**

A) To prevent over-revving the engine when downshifting

B) To synchronize the left and right drive axles during turns

C) To prevent the clutch from engaging while the vehicle is moving

D) To match gear speed to shaft speed before engagement

Q48. **When checking a driveshaft working angle, what is the maximum acceptable operating angle for most heavy truck driveshafts to prevent vibration and U-joint wear?**

A) 12 degrees

B) 3 degrees

C) 7 degrees

D) 1 degree

Q49. **What does "HPFP" stand for in a common-rail diesel fuel system, and what is its function?**

A) Hydraulic Pump Fuel Pressure — refers to fuel pressure at the hydraulic pump feed

B) High-Performance Fuel Proportioner — balances fuel delivery to all cylinders

C) High-Pressure Fuel Pump — pressurizes fuel to common rail pressure

D) Hot-Phase Fuel Processor — heats fuel for cold-start performance

Q50. **What is the minimum tread depth for tires on an active steering axle of a commercial vehicle under National Safety Code Standard 11?**

A) 2 mm, the same figure that applies at every other axle

B) 3 mm, deeper than the figure for other axle positions

C) There is no minimum — tires are replaced on driver judgment

D) 6.4 mm, because a steering axle needs deep tread for stability

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 161-question 310T bank; the last page shows what the complete printable bank contains.

Q1. Answer: D

Late/high-RPM shifts: TCM receiving a high throttle demand signal. The Transmission Control Module uses throttle position (or percent load) to determine shift points. If the TPS reads higher than actual (or the driver demand signal is incorrectly high), the TCM interprets it as high driver demand and delays upshifts to maintain "performance mode." Diagnose: monitor TPS on a scan tool at steady throttle. Also check for a stuck throttle plate or accelerator pedal sensor.

Q2. Answer: A

Light heat checking with material still available: recondition the flywheel, then verify thickness and runout. A flywheel has two outcomes at clutch replacement — recondition it or replace it — and the deciding question is whether the damage can be cut away while the face still finishes above the minimum thickness marked on the flywheel. Light checking and shallow grooving sit inside that. Installing a new disc on the face as found transfers the damage pattern straight into the new lining, and neither adhesive nor hand sanding removes the damaged layer or restores a flat face. The other side of the same rule: a flywheel whose cracks run deep or out to the edge, or one already sitting at minimum thickness, has no reconditioning path left and is replaced.

Q3. Answer: A

More than 3 PSI per minute with the brakes released is excessive leakage under every Canadian standard — find it and repair it. On periodic inspection, NSC Standard 11 Part B Section 3A rejects a truck whose air pressure drops more than 7 kPa (1 psi) per minute during the leakage test, and lists a drop of more than 20 kPa (3 psi) per minute as a Hazardous Condition; a trailer is rejected above 28 kPa (4 psi) per minute when attached to a towing vehicle, or above 20 kPa (3 psi) per minute on a shop air source, and is hazardous above 40 kPa (6 psi) per minute. The driver-level test in the Ontario Air Brake Handbook is done with the engine off and the service brakes fully applied, allowing 21 kPa (3 psi) per minute for a straight truck, tractor or bus, 28 kPa (4 psi) for a tractor and trailer, and 41 kPa (6 psi) for a tractor with two or more trailers. Leakage with the brakes released should be lower than with them applied, so this reading fails all of them. Build to governor cut-out, shut the engine off, time the drop, then hunt the leak with soapy water.

Q4. Answer: D

Chatter that appears only in slow tight turns is a differential that cannot absorb the speed difference. On a tandem tractor the first thing to check is the inter-axle differential (power divider) lock: a dash valve left engaged, a leaking or blocked air line, a sticking lockout cylinder, or a bent shift fork will hold the two drive axles together, and the tandem then binds and releases through every tight turn. Only after the lock is confirmed fully disengaged does the lubricant come into it, and then only on a limited-slip unit, where the wrong oil or a depleted friction modifier makes the clutch pack grab and slip. Worn universal joints, loose mounts and worn steering linkage produce symptoms that do not switch on and off with steering angle.

Q5. Answer: D

Reject at more than 7 kPa (1 psi) per minute with the brakes released. NSC Standard 11 (January 2020), Part B Section 3A, item 2 'Air Supply System', condition (g), rejects a truck whose air pressure drops at more than 7 kPa (1 psi) per minute during the leakage test. A drop of more than 20 kPa (3 psi) per minute is that section's hazardous-condition (out-of-service) tier, not the acceptance limit. Condition (f) of the same item separately rejects a loss of more than 138 kPa (20 psi) during a full service application. The driver-level applied test uses different figures: with the brakes held fully applied, the Ontario Official Air Brake Handbook allows 21 kPa (3 psi) per minute for a straight truck, tractor or bus, 28 kPa (4 psi) per minute for a tractor-trailer, and 41 kPa (6 psi) per minute for a tractor with two or more trailers.

Q6. Answer: B

Speed-specific floor/seat vibration = likely driveshaft balance or phasing issue. Driveshaft vibration is proportional to rotational speed (square of speed for imbalance). At low speeds, even significant imbalance produces tolerable vibration. Above a threshold speed, vibration becomes severe. A phasing error — the yokes at each end of the shaft not aligned in the same plane — leaves the two joints' speed fluctuations adding instead of cancelling, which creates a second-order vibration. Front imbalance vibrates through the steering column/wheel. Rear driveshaft vibration transmits through the cab floor. Confirm with a driveshaft balance check and universal joint phasing inspection.

Q7. Answer: D

Good supply voltage plus dim lamps means the voltage is being lost on the way there. Corroded grounds, loose connectors and oxidized fuse holders drop voltage under load, and whatever is dropped across them never reaches the bulbs — which is also why a single shared ground explains both lamps dimming together, where a wrong bulb would normally affect only the one it was fitted to. Confirm it with a voltage drop test rather than a resistance check: with the lights on, put the meter across each connection, splice, switch and ground in turn, working both the feed side and the return side. A healthy segment drops almost nothing; the segment carrying most of the drop is the fault. Compare the readings against the manufacturer's service information for that circuit.

Q8. Answer: B

A bulge up to 9 mm high caused by a sidewall repair is accepted; what fails a tire is a bulge caused by separation. National Safety Code Standard 11, Part B, Section 9 Item 3 covers the tire sidewall and manufacturer markings. Its condition criteria reject a tire that has a bump or bulge caused by tread or sidewall separation, and the same item carries a note that a bulge of up to 9 mm in height due to a sidewall repair is acceptable, and that such a bulge may sometimes be identified by a blue triangular label in the immediate vicinity. Item 3 then lists a visible bump or bulge in the sidewall area greater than 9 mm in height as a hazardous condition, so height is what separates an accepted repair from a condition that keeps the vehicle in the shop. Read the two together and the rule is that separation fails the tire, and so does a repair bulge past 9 mm, but a repaired sidewall on its own does not. Item 3 does reject a plug-type repair, or a rubber-coated or cured rubber plug used in a sidewall, which is why a plug is worth looking for; a blue triangular label marks the accepted repair, not a plug. Ply separation, exposed body cords, a broken or distorted casing and UV degradation more than 3 mm deep are the other reject conditions in the same item.

Q9. Answer: C

Above 350 mm nominal, the fallback limit is 3.0 mm over the original diameter. National Safety Code Standard 11 Part B, Periodic Motor Vehicle Inspection, Section 3A — Air Brakes, rejects a drum whose measured diameter exceeds the limit indicated on the drum, the OEM standard, or an industry standard. Only when none of those is available does the standard supply its own figures: 2.3 mm over original diameter for a nominal drum size of 350 mm (14 in) or less, and 3.0 mm over original for a nominal size greater than 350 mm. A 420 mm drum falls in the larger group, so 3.0 mm applies. The order matters as much as the number — the marking on the drum comes first, and these figures are the last resort.

Q10. Answer: D

A friction modifier. A limited-slip differential adds a clutch pack or a cone clutch between the side gears and the differential case, so that when one wheel starts to spin some torque is still carried across to the wheel that still has grip. Those clutches are meant to slip a little and grip a little every time the axle goes round a corner, because the two wheels are travelling different distances and the differential has to let them. Plain hypoid gear oil makes them grab and release in steps rather than slipping smoothly, and the axle chatters and shudders through slow tight turns while the clutch faces glaze. The friction modifier is an additive that narrows the gap between static and sliding friction in the clutch pack so the slip is smooth and quiet. Some gear oils are sold already modified for limited-slip service; otherwise the modifier is added separately in the quantity the axle maker states. The other three answers describe choices made for other reasons: viscosity and the gear-oil rating are chosen for the ring and pinion and do not change because the differential is limited-slip, and a detergent package belongs to engine oil, which has combustion by-products to carry, not to axle oil.

Q11. Answer: A

Pressure low at idle, OK at RPM = worn bearings. Worn main or rod bearings with excessive clearance allow oil to escape faster than the pump supplies at low RPM; at high RPM, pump output overcomes the leakage. Compare oil pressure to spec: typically 10 PSI per 1000 RPM minimum.

Q12. Answer: A

At cut-out the governor pilots reservoir air down the unloader line to the compressor, and that air holds the compressor's inlet valves open. The compressor keeps turning, but it simply shuttles air back and forth instead of pumping it into the reservoirs. A truck air compressor is driven off the engine gear train (or a belt) and turns the whole time the engine runs, so nothing disengages it and there is no shutoff valve in the discharge line. When reservoir pressure falls to cut-in, the governor exhausts the unloader line, the inlet valves close, and the compressor loads again. The reservoir safety valve is a last-resort pop-off, not the normal pressure control - if it is lifting, something in the governor or unloader circuit has already failed. This is why a compressor that will not stop building, or one that never builds at all, is diagnosed by testing the governor and the small line running to the compressor unloader before the compressor itself is condemned.

Q13. Answer: A

DEF quality and concentration are the first thing to suspect when SCR conversion efficiency drops. Selective catalytic reduction uses diesel exhaust fluid, a 32.5 percent urea solution, to convert NOx to nitrogen and water. Diluted, contaminated, or frozen-then-improperly-thawed fluid degrades that conversion, and the downstream NOx sensor sees the result. The same fault family is triggered by low fluid level, a dosing injector fault, a fluid heater fault, and a degraded catalyst. A loaded particulate filter does not raise downstream NOx, and an EGR valve stuck open lowers NOx rather than raising it. Verify fluid quality and level before condemning the sensor or the catalyst.

Q14. Answer: C

The modulator is an on/off air valve with two solenoids (hold and exhaust) controlled by the ABS ECU. When the ECU detects impending lockup it energizes both solenoids so chamber air **exhausts** through the modulator exhaust port; with only the hold solenoid energized it **holds**, blocking supply air and sealing the exhaust; with both de-energized it **reapplies**, passing air to the chambers as in normal braking. It does not read wheel rotation - the wheel speed sensors feed that information to the ECU, which makes the lockup decision. It is not a load-proportioning device, and its delivery port feeds the service actuators, so it never applies the spring brakes.

Q15. Answer: C

Battery discharge through "off" switch = switch not isolating. Possible causes: corroded or worn contacts in the isolator switch that no longer make a complete break, or a bypass circuit that bypasses the switch (alarm system, GPS tracker, fridge). With the switch OFF, measure the current with an ammeter in series between one battery terminal and everything connected to it, as for a key-off drain test, so that the meter sees the whole draw, including any circuit wired around the switch. A clamp-on ammeter works only if it is a DC (Hall effect) clamp that resolves milliamps: a current transformer clamp does not read DC at all, and a clamp built for large currents may not register a draw below about half an ampere.

Q16. Answer: D

A documented full inspection with the drums removed qualifies a truck's drum brakes for a limited inspection for 19 months. National Safety Code Standard 11 Part B, Periodic Commercial Motor Vehicle Inspections, opens Section 3 — Brake Systems with a part headed Options for Inspecting Internal Brake Components. It names three drum brake inspections: a full inspection with the drum removed, a wheel-on full inspection available only on cam-type brakes with removable dust shields, and a limited inspection, which the standard describes as an inspection through the inspection holes involving a measurement of shoe lining only. When the full inspection with the drum removed is done on a truck or trailer with drum brakes and the documentation is completed, that brake qualifies for the limited inspection for 19 months. A bus with drum brakes earns only 7 months on the same evidence. Disc brakes are on their own clock: they need a full inspection with the wheels removed at least every 12 months, and 7 months of limited inspection follows it. Two limits sit on the exemption. It covers the internal brake components alone, so the vehicle still goes through its periodic inspection on its normal cycle. And it is forfeited wheel end by wheel end: where a defect is found or suspected, that brake no longer qualifies for a wheel-on full inspection or a limited inspection, and the drum comes off.

Q17. Answer: B

No material left to cut, and cracks that run into the casting: the flywheel is replaced. Reconditioning and replacement are the two outcomes for a flywheel, and reconditioning is only available while the damage can be machined away and the face still finishes above the minimum thickness marked on the flywheel. This face is already at that limit, so any cut takes it under spec, and cracks that reach the outer edge extend into the casting rather than sitting in the surface layer — a grinder would leave them there. Torching does not close a crack in cast iron, and dressing the crack edges leaves both the cracks and an out-of-flat face. Compare the light heat-checking case, where material is still available and the flywheel is reconditioned instead.

Q18. Answer: D

DEF quality faults: wrong concentration, contamination, or sensor failure. DEF is a nominally 32.5% urea solution (AUS 32 / ISO 22241) held to a narrow concentration band. Diluted DEF (too much water) reduces SCR efficiency and trips the sensor. Contaminated DEF (wrong fluid added to the tank) can damage the SCR catalyst. The DEF quality sensor measures urea concentration using sound wave velocity or electrical properties. A faulty sensor can trigger quality faults even with good DEF. Test: compare a refractometer reading of the DEF to specification. DEF that froze and thawed incorrectly can also trigger the fault.

Q19. Answer: A

A 175 PSI cut-out drives the supply side past what the dryer, valves, fittings and reservoirs are built to hold. The Official Air Brake Handbook sets the ceiling plainly: actual cut-out pressure must never be higher than 1000 kPa (145 psi), and cut-out normally sits only 138 to 173 kPa (20 to 25 psi) above cut-in. Safety valves normally open at 1035 kPa (150 psi), so a 175 PSI setting leaves the reservoir safety valve venting continuously — and a safety valve that is venting air means too much pressure is building in the system and needs immediate repair. Seals, hoses, the dryer purge valve and fittings all work above their design pressure, and the compressor runs loaded far longer than intended. The extra pressure buys nothing at the brakes: application pressure is metered by the treadle and relay valves, and the spring brakes are held off by air but applied by their springs, so parking hold does not change. Reset the governor to the OEM cut-in and cut-out figures with a calibrated gauge.

Q20. Answer: A

NSC Standard 11 Part B: a full periodic inspection at least every 12 months, performed by a certified inspector. A commercial vehicle undergoes a full periodic inspection at a designated facility at least once every 12 months, and some jurisdictions set a shorter cycle for particular vehicle types. The inspection covers all major safety systems: brakes, tires, lights, steering, suspension, frame, and more. A decal or certificate is issued after the vehicle passes. The two verdicts the standard gives the inspector are reject, meaning the vehicle fails and no decal is issued until the condition is corrected, and hazardous condition, meaning the vehicle is too unsafe to be driven until it is corrected. Operating a commercial vehicle whose periodic inspection has expired is an offence under the regulation of the jurisdiction that adopted the standard. Out of service is a separate instrument altogether: it is the roadside criteria applied by an enforcement officer, and it is not something a periodic inspection issues.

Q21. Answer: A

White smoke + coolant loss + rough idle = coolant entering combustion (failed head gasket). Coolant burns as white steam with a sweet smell. Combustion pressure also enters the cooling system, pressurizing it (visible bubbles in the coolant reservoir). A head gasket failing between a coolant passage and a cylinder allows coolant to flood that cylinder — a hydrostatic lock risk if left unaddressed. Confirm with a combustion leak test (CO detector at reservoir).

Q22. Answer: D

Rated RPM is the full-load governed speed — the RPM at which the engine makes its advertised maximum horsepower. Diesel torque peaks lower, around 1,100–1,500 RPM on many truck engines, while horsepower peaks at rated speed, around 1,800–2,100 RPM. Rated speed is not the engine's top speed: with the load removed, the governor lets the engine run up to high idle, the no-load governed speed, which sits above rated speed, and the over-speed limit sits above that again. Operating at rated RPM under load represents maximum power output from the engine.

Q23. Answer: B

Terminals 85 and 86 are the coil; 30 and 87 are the load contacts. Under the DIN 72552 terminal designations used on ISO mini relays, the switch energizes the coil through terminals 85 and 86 with a small control current - typically about 50 to 200 milliamps - and the magnetic field closes the contacts between terminal 30 (battery feed) and terminal 87 (switched output to the load). That split is the whole point of a relay: the dash switch and its thin wiring carry only the coil current, while the relay's contacts carry the heavy load current on a short, fat run near the battery. Quick bench check: energize 85 and 86, listen for the click, then verify continuity from 30 to 87 with the coil energized and no continuity with it released.

Q24. Answer: C

Oscilloscope: captures the waveform over time — reveals intermittent faults. A DMM reads average/RMS voltage and misses brief dropouts. An oscilloscope displays the actual PWM waveform — a dropout, missing pulse, or duty cycle change appears instantly on screen. Essential for intermittent electronic faults on PWM-controlled circuits.

Q25. Answer: B

Clutch slipping: oil/grease contamination on friction surface — pressure plate cannot grip the disc. Oil from a leaking rear main seal or improperly lubricated parts contaminates the disc friction material. Oil-soaked friction material has greatly reduced coefficient of friction. Other causes: worn friction disc (too thin, past minimum thickness), weak or broken pressure plate springs, overheated pressure plate (springs lose tension from clutch abuse). Contamination: replace both disc and pressure plate — cleaning oil from friction material is ineffective.

Q26. Answer: A

Normal in this context: an automatic adjuster takes up clearance a little at a time over a series of applications, not in one. The adjuster only advances when the stroke reaches its adjustment threshold, so after a deliberate back-off during installation it may need several full applications at operating pressure before it has ratcheted back to the correct clearance. Run the number of applications the manufacturer's procedure calls for, then measure the stroke. Do not condemn a new unit for failing to recover on the first pull, and do not read the delay as a broken pawl. Two limits sit on this answer. First, it describes the installation and verification procedure, where backing off is part of the published method — it is not a description of what to do to a truck in service. Second, if the stroke is still outside the limit once the procedure is complete, the unit is not slow, it is faulty, and the correct response is repair or replacement of the adjuster or of whatever in the foundation brake is defeating it. That distinction matters because an automatic adjuster found at or beyond its stroke limit in service must never be wound back by hand to make it pass: National Safety Code Standard 11 states that such a brake requires repairs and that a manual adjustment will not correct the problem.

Q27. Answer: C

EGR stuck open at cruise: rough running, loss of power, excessive exhaust smoke. At highway speeds, the engine needs maximum fresh air for full-load combustion. EGR gases are depleted of oxygen — introducing them at cruise reduces the effective air charge. The combustion is oxygen-starved, causing rich combustion (black smoke), loss of power, and rough running. The EGR is typically commanded closed at high load/high RPM by the ECM — a stuck-open mechanical valve overrides this. The engine may also misfire.

Q28. Answer: D

Intercooler can hold pressure but have poor heat transfer — bent fins, oil contamination, or internal blockage. A pressure leak test confirms no external air loss, but does not confirm thermal efficiency. If the intercooler core fins are bent (reducing airflow through the matrix), coated with oil from turbo leakage (oil acts as insulation), or internally scaled, heat transfer is reduced — outlet temperature remains high. Measure inlet and outlet air temperatures: large Δt = good. Also check ambient air restriction to the intercooler.

Q29. Answer: A

On a turbocharged engine, manifold absolute pressure under full load must be well above atmospheric. 98 kPa under full load means no boost. At key-on with the engine stopped, the MAP sensor should read barometric pressure, about 101 kPa at sea level — that reading confirms the sensor and its circuit are alive and correctly referenced. Once the engine is loaded, a working turbocharger raises manifold pressure well above barometric. A reading that stays at or just below barometric means the charge air system is building nothing: a leaking charge air pipe or intercooler, a failed turbocharger, or a wastegate stuck open. An intake restriction such as a plugged air filter or a collapsed intake hose holds manifold pressure below barometric as well. A naturally aspirated engine behaves this way normally, which is why the engine type must be established before the reading can be judged.

Q30. Answer: D

An amber trailer ABS lamp is a defect to be recorded and repaired, not a condition that takes the vehicle off the road. A trailer-mounted lamp that stays on after the bulb-check cycle is a reject at a periodic inspection under National Safety Code Standard 11, and the fault must be recorded on the driver's inspection report and repaired. It is not a hazardous condition on its own: Standard 11 reserves that label for any malfunction of the ABS that prevents normal brake operation. With the ABS inactive the braking system reverts to conventional operation, so the trailer still stops under a service application. A carrier that knowingly leaves the fault unrepaired is in violation.

Q31. Answer: D

PTC: resistance rises with temperature. NTC: resistance falls with temperature. NTC thermistors are used for temperature sensing: as temperature rises, resistance drops, changing the voltage at the ECM input. The ECM interprets this voltage as temperature. PTC thermistors are used as circuit protectors (self-resetting current limiters) and as heater elements (glow plugs in some designs). Both types are passive resistors — they require an applied voltage to measure or limit current.

Q32. Answer: B

Two-way check valve: passes the higher of two inlet pressures to its outlet. Commonly used in spring brake circuits: either the service brake signal OR the parking brake supply can apply the spring brakes. The two-way check valve ensures that whichever signal is higher (emergency application OR service application) is what moves the spring brake piston. This allows the service brakes and parking brakes to share the same spring brake chambers without interfering with each other.

Q33. Answer: D

Under 5 mm at any point on a continuous strip lining is a reject. National Safety Code Standard 11 Part B, the mandatory Periodic Motor Vehicle Inspection, Section 3A — Air Brakes, rejects a bonded or riveted continuous strip brake shoe lining thinner than 5 mm at any point, and a bolted or riveted block type lining thinner than 8 mm at any point. The standard is explicit about where to measure: minimum allowable thickness is judged at the thinnest point of the lining, while the figure written on the inspection report is taken at the edge of the lining near the centre of the shoe, on the thinner of the two shoes. Vehicles with hydraulic or electric brakes are a separate case — Section 3H rejects bonded lining under 2 mm and bolted or riveted lining under 3 mm. The 1/16 inch (1.6 mm) figure is nowhere near what a Canadian periodic inspection will pass on an air brake.

Q34. Answer: D

Never increase fuse rating as a fix. The fuse protects the wiring, not the component. A blown fuse means the circuit is drawing more current than the wire can safely carry. Increasing the rating allows wiring to overheat and potentially start a fire. Diagnose why the circuit draws excess current — short circuit, overloaded component, damaged wiring.

Q35. Answer: B

Everything between the air cleaner and the turbocharger inlet runs under suction, so a leak there feeds the engine unfiltered air. Airborne dust and sand are abrasive. Once past the filter they scour the compressor wheel, then the cylinder walls, rings and bearings, and that wear is permanent — no amount of later filtration undoes it. A dust track streaking away from a clamp, a split hose or a missing gasket on the clean side of the filter is the classic sign. Note which way the other choices have the air moving: this length of pipe is under vacuum, so air is drawn in rather than blown out, and nothing on the intake side raises exhaust back pressure at the turbine.

Q36. Answer: C

Bobtail = drive axle nearly unloaded, so the system trims rear brake pressure to match. With no trailer, a tractor's drive axle carries a small fraction of the weight it was braked for, and full rear application pressure would slide the wheels on all but the driest pavement. The bobtail proportioning system senses the no-trailer condition and delivers reduced pressure to the drive-axle chambers during service applications, keeping brake force in proportion to the load on the axle; normal delivery returns when a trailer is coupled. It does not shut the rear brakes off, nothing raises delivery above what the treadle meters, and trailer-circuit air is not redirected anywhere. The spring brake (park and emergency) function is mechanical and unaffected. Diagnostic angle: low delivery pressure at the drive axle on a bobtail tractor can be this system working as designed - confirm the proportioning function before condemning foundation parts for 'weak rear brakes'.

Q37. Answer: A

A broken power spring cannot develop the clamping force the parking brake depends on. A spring brake applies mechanically: exhausting the hold-off air lets a compressed coil spring drive the pushrod out. A fatigued, cracked or broken spring still applies the brake, but with too little force to hold the vehicle — and a grade is what exposes it. The stem rules out the usual rivals: full reservoirs mean supply is fine, and in-limit stroke means the foundation brake is adjusted. A relay valve that failed to release service air would leave the brakes dragging rather than free, closing the tractor protection valve applies the trailer spring brakes rather than releasing them, and the low pressure warning switch only drives a lamp and buzzer — nothing signals a spring brake to apply, because the application is what happens when air is taken away.

Q38. Answer: B

GL-4 for manual transmissions (brass synchronizers), GL-5 for differentials. GL-5 gear oil contains high concentrations of extreme pressure (EP) additives — typically sulfur-phosphorus compounds — that are activated under high loads in hypoid gear sets (differentials). These additives are chemically aggressive to brass and bronze, which are used in synchronizer rings in manual transmissions. Brass synchronizer corrosion from GL-5 causes sluggish, difficult shifting. Always use the OEM-specified oil type. Note: some manual transmissions specify ATF or a transmission-specific fluid.

Q39. Answer: D

Quick-release valve: rapid local air exhaust from brake chambers on release, allowing spring return and reducing brake drag. Without a quick-release valve, air would have to travel all the way back through the treadle valve and exhaust at the driver's feet — taking too long and causing brake drag. The QR valve is located close to the brake chambers: when the pedal is released and line pressure drops, it immediately exhausts air locally at the chamber — rapid piston return and immediate brake release.

Q40. Answer: D

A correctly charging 12V alternator should produce 13.8–14.5V. 12.1V is battery surface voltage — not charging voltage. This indicates the alternator is not producing output, or output is below the battery voltage (not charging). Causes: failed alternator, broken belt, failed voltage regulator, open field circuit, poor ground connection. Left undiagnosed, the battery will drain and the vehicle will experience increasingly severe electrical problems.

Q41. Answer: A

Oil cooler bypass valve: opens on cold thick oil. Cold engine oil has high viscosity and creates a high pressure differential across the oil cooler. The bypass valve opens to route oil around the cooler, ensuring adequate lubrication flow during cold starts before oil warms and thins. Once warm, the valve closes and oil flows through the cooler normally.

Q42. Answer: B

The Transportation of Dangerous Goods Act, 1992 and its Regulations set three duties that travel together. Anyone who handles, offers for transport or transports dangerous goods must be adequately trained and hold a valid training certificate, or do the work in the presence and under the direct supervision of someone who is — so training does not stop at the shipper. A shipping document must accompany the shipment, and the vehicle must display the correct safety marks for the class and quantity being carried. An Emergency Response Assistance Plan is required for certain higher-risk goods. There is no dangerous goods endorsement on a Canadian driver's licence; the training certificate is the credential.

Q43. Answer: A

Relay valve: speeds up brake application at remote axles. Without a relay valve, air must travel the full length of the vehicle (foot valve to rear brakes). The relay valve uses a small pilot signal from the foot valve to open a local valve, delivering air from a nearby reservoir rather than the foot valve — much faster response.

Q44. Answer: C

Correct alternator voltage but poor charging: check current flow and circuit resistance. 14.2V is within spec, but if the charging circuit has high resistance (corroded battery cable or charge wire, poor battery terminal connection), current flow to the battery is restricted despite correct voltage output. A battery with high internal resistance also won't accept charge efficiently. Test: measure charging current (clamp ammeter), measure voltage drop in the charging circuit, load test the battery.

Q45. Answer: C

Every defect goes on the trip inspection report, and a major defect stops the vehicle. In Canada the driver inspects the vehicle against National Safety Code Standard 13 Schedule 1 and completes a trip inspection report; the inspection covers the following 24 hours, and the report is carried and produced on demand. Where no defects are found, the report records that fact. A defect found at the inspection, or detected later in the day while driving, is entered on the report and reported to the carrier before the next required inspection. A major defect is reported to the carrier immediately and the vehicle must not be driven until it is corrected. A defect that is not a major defect is recorded and the carrier has it corrected before the next required inspection, so the driver may continue once it is written down. The carrier keeps the report for at least 6 months. One point of vocabulary: Standard 13 calls this the trip inspection and its record the inspection report. The initials DVIR are heard in Canadian shops but come from the American scheme and are not the term in the standard.

Q46. Answer: D

12V is standard for most North American trucks. European trucks commonly use 24V. Some newer hybrid systems use 48V. Two 12V batteries wired in parallel = 12V (more capacity). Two wired in series = 24V (used in European and some heavy equipment).

Q47. Answer: D

A synchronizer brings the gear being engaged and the shaft to the same speed before the dog teeth are allowed to lock. It is a small cone clutch. As the collar slides toward the gear it first pushes a friction ring onto a matching cone on the gear, and that friction drags the two to a common speed. A blocker ring holds the collar back until the speeds match, then indexes out of the way so the teeth engage without clashing. Worn rings and cones lose both the friction and the blocking, and the driver has to match speeds by hand, double-clutching or rev-matching, to shift cleanly. Note where this applies. The main section of a Class 8 transmission is normally non-synchronized: its sliding clutches engage the gear's clutching teeth directly and the driver does the speed matching, which is why double-clutching is the technique on that equipment and why a worn blocker ring cannot be the diagnosis on such a box. Where a synchronizer does appear in a heavy transmission it is usually in the auxiliary range section rather than in the main box.

Q48. Answer: B

Maximum continuous operating angle: approximately 3 degrees. U-joints run at large angles generate torsional excitation and wear rapidly; higher angles are tolerated only at lower shaft speeds. Two companion rules: the operating angles at the two ends of the shaft must be equal within 1 degree, and every joint needs at least 1 degree of angle so its needle rollers keep rotating instead of brinelling the cross. Cancellation of the speed fluctuation requires BOTH conditions met at once - equal angles at the two joints AND correct phasing, meaning the in-board yokes of the shaft in line with each other. A shaft that is out of phase, or whose joint angles are unequal, leaves the second joint adding to the first joint's fluctuation instead of cancelling it. Check phasing and measure the angles as separate steps of a driveline inspection, and remember ride height changes the angles.

Q49. Answer: C

HPFP = High-Pressure Fuel Pump. In a common-rail system, the HPFP takes fuel from the low-pressure supply delivered by the feed (transfer) pump and raises it to rail pressure. How high the system can go depends on its design: Bosch's modular common-rail system for commercial vehicles is built for system pressures of up to 2,500 bar, in versions scalable between 1,800 and 2,500 bar, and its CPN6 heavy-duty pump handles pressure levels from 1,800 to 2,500 bar. This high-pressure fuel is stored in the common rail and distributed to solenoid or piezo injectors. The HPFP is driven by the engine and is the heart of the common-rail injection system.

Q50. Answer: B

A tire on an active steering axle is rejected below 3 mm of tread; every other tire is rejected below 2 mm. National Safety Code Standard 11 sets the steer figure higher because steering control and wet-weather braking depend on the front tires. Below the reject figure the vehicle fails a periodic inspection; the hazardous condition that takes it off the road is less than 2 mm on a front tire and less than 1 mm on a rear tire. Measure at a major tread groove with a tread depth gauge, never on a wear bar, and record the lowest reading on the inspection report. Some jurisdictions require 3 mm on all tires of certain vehicles, and vehicles carrying dangerous goods may face a different figure, so check the applicable provincial requirement. The American tread depth figures are different and do not set the requirement in Canada.

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